METANOIA

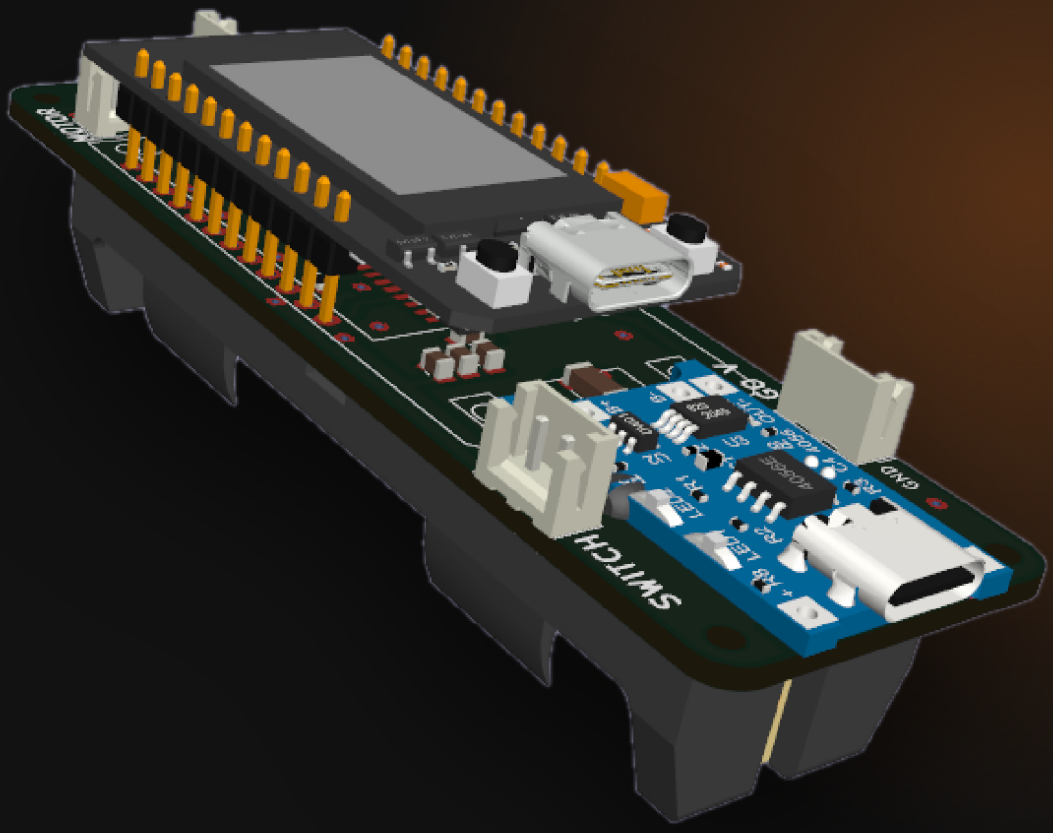
TECHNICAL DOCUMENTATION · REV A

ESP32 · BATTERY POWERED · MOTION CONTROL

Motor Control Carrier Board

Bidirectional DC-motor speed and direction control with live voltage and current telemetry, a switched LED output, on-board Li-ion charging and reverse-polarity protection.

TTGO T-Display | T-Display-S3 | MX1508 | INA219 | TP4056 | 18650



PROJECT	REVISION	TOOLCHAIN	DATE
metanoia.kicad_sch	Rev A · design review	KiCad 10.0.1	30 September 2026

DOCUMENT

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Document information

BOARD	Motor Control Carrier Board (carrier for LilyGO TTGO T-Display / T-Display-S3)
SOURCE	`metanoia.kicad_sch` — single sheet, KiCad E.D.A. 10.0.1
STATUS	`Rev A · design review` Three blocking findings — see section 11 before ordering PCBs
PREPARED BY	Metanoia for Information and Technology — Helwan, Cairo

01 INTRODUCTION Overview

The Motor Control Carrier Board turns a LilyGO display module into a self-contained, battery-powered motion controller. It drives one brushed DC motor in both directions with PWM speed control, measures the motor's supply voltage and current in real time, switches an external LED load, and reads a two-channel encoder — all from a single 18650 cell that charges over USB-C.

The controller socket accepts either the **TTGO T-Display (ESP32)** or the **T-Display-S3 (ESP32-S3)**. Both modules share the same header footprint; only the GPIO numbers behind each pin differ, which firmware resolves at compile time (section 09).

3.7 V	≈1.5 A	1 mA	1 A
1S Li-ion, 3.0–4.2 V	Motor channel, continuous*	Current-sense resolution	USB-C charge rate

Features

M — Motor control
MX1508 H-bridge: forward, reverse, coast and brake with PWM speed.

V·I — Telemetry
INA219 on a 10 mΩ high-side shunt reports motor voltage, current and power over I²C.

L — LED output
Low-side MOSFET channel for an external LED load, PWM dimmable.

E — Encoder input
Two-channel 3.3 V input on J4 for speed and position feedback.

+ — Charging
TP4056 module with USB-C input and cell protection circuitry.

⇄ — Protection
Reverse-polarity stage built from two AO3401A P-MOSFETs in parallel.

Key specifications

PARAMETER	VALUE
Supply	1× 18650 Li-ion cell in holder BT1 (Keystone 1042P), 3.0–4.2 V
Charging	USB-C on TP4056 module, 1 A (module default)
Controller	LilyGO TTGO T-Display (ESP32) or T-Display-S3 (ESP32-S3), drop-in
Motor output	1× brushed DC, MX1508 channel 1, connector J3
Sensing	INA219, 10 mΩ shunt, I²C address `0x40`, up to 3.27 A reported
Logic rail	+3.3 V supplied by the controller module's regulator
Power control	External on/off switch on J2, in series with the battery output

* Typical MX1508 rating per channel; confirm against the driver datasheet and the motor's stall current.

HOW IT FITS TOGETHER

02 System architecture

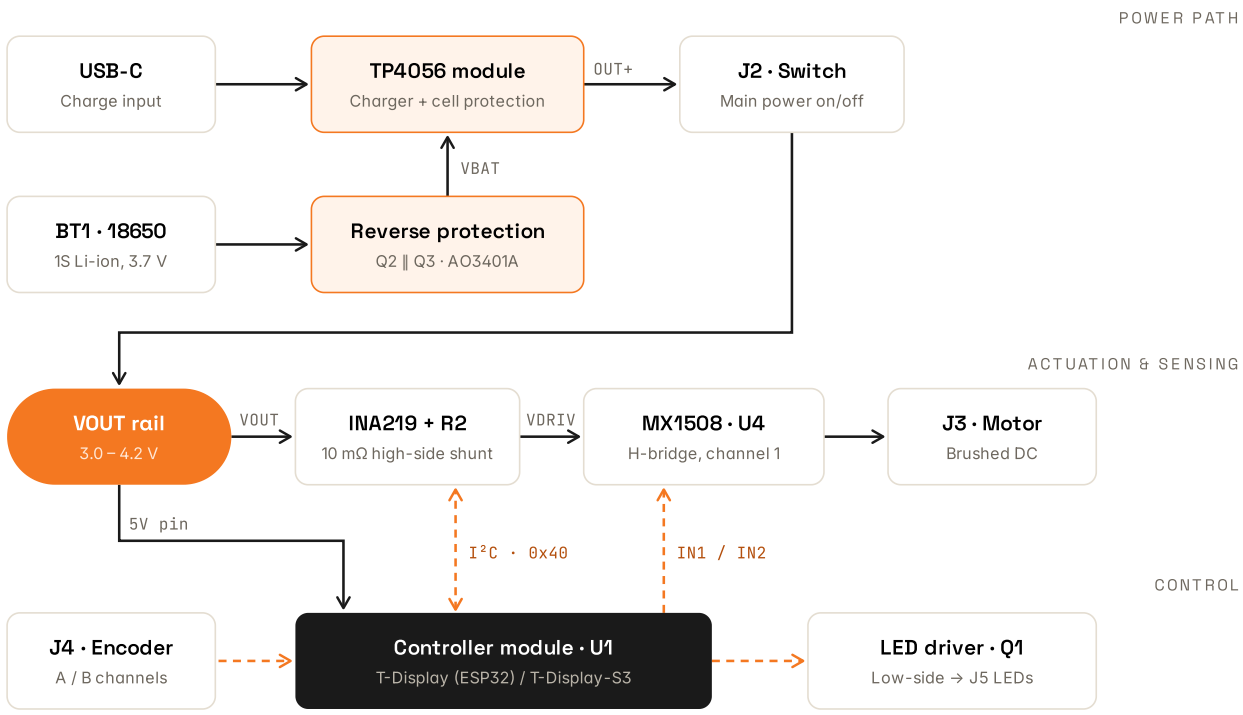


FIG 2.1 Functional block diagram. Solid lines carry power; dashed orange lines carry control and data.

Power rails

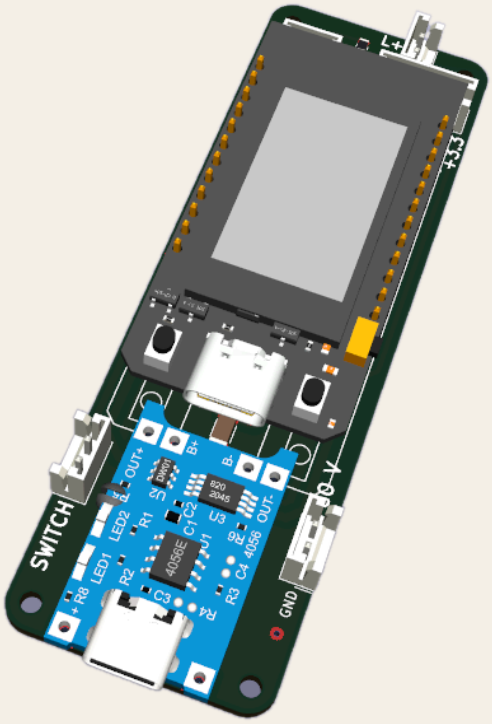
NET	RANGE	SOURCE	FEEDS
VBAT	3.0 – 4.2 V	18650 cell via Q2 ‖ Q3	TP4056 module `B+`
VOUT	3.0 – 4.2 V	Module `OUT+` through switch J2	Controller 5V pin, J1, J5 LED supply, shunt R2
VDRIV	VOUT − I·10 mΩ	Load side of shunt R2	MX1508 `VDD1/VDD2` (motor supply)
+3.3V	3.3 V	Controller module regulator (3V pins)	MX1508 logic, INA219, I²C pull-ups, encoder

Signal paths

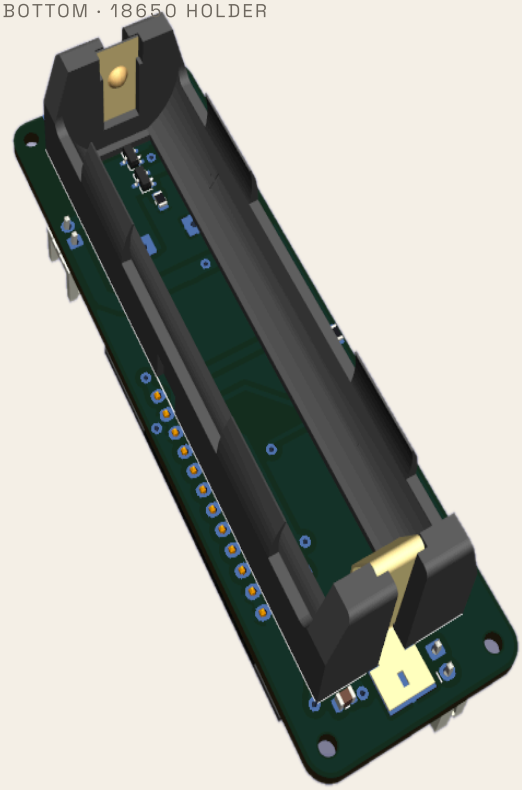
The controller drives the H-bridge inputs directly with PWM, reads the INA219 over I²C (4.7 kΩ pull-ups R4/R5 on board), switches the LED channel through Q1 and samples the encoder on J4. Because the shunt sits between `VOUT` and `VDRIV`, the INA219 sees **only the motor current** — the controller and LED loads are excluded from the measurement.

03 BOARD RENDERS
Hardware views

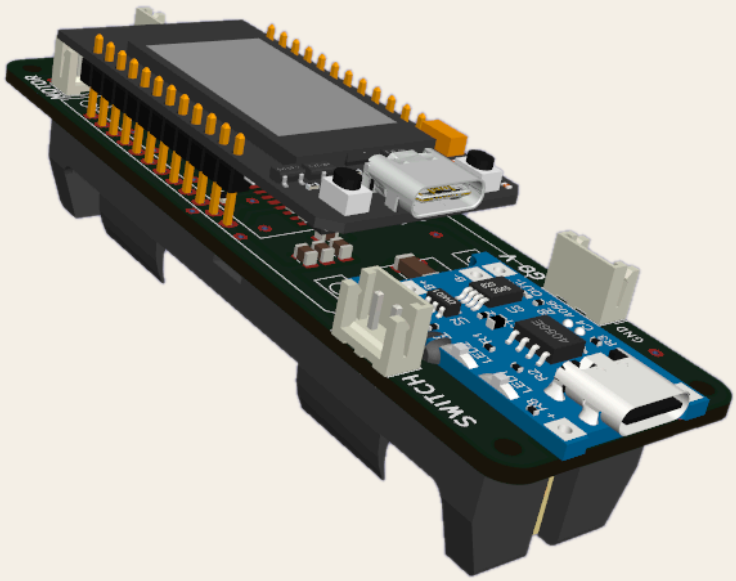
TOP · ASSEMBLED



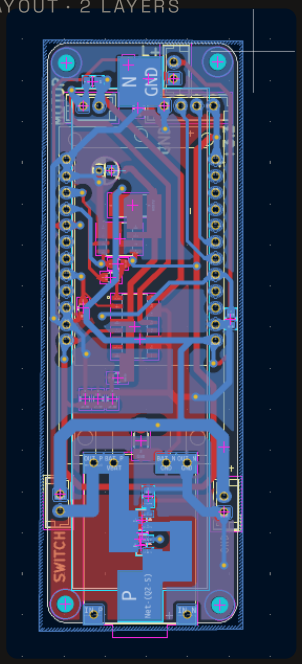
BOTTOM · 18650 HOLDER



ISOMETRIC



PCB LAYOUT · 2 LAYERS



The controller module sits on the top side over the header rows; the TP4056 charger module is mounted at the lower edge with its USB-C port facing out. The 18650 holder occupies the full underside, keeping the footprint to the size of the cell. External connections — motor, switch, LEDs, encoder and module VIN — use 2.54 mm JST-style headers along the board edges.

04 Power subsystem

BATTERY, CHARGING, SWITCHING

A single 18650 cell powers the entire board. The cell passes through the reverse-polarity stage onto `VBAT`, which connects to the TP4056 module's battery terminal. The module's protected output feeds the main switch J2, producing the `VOUT` rail.

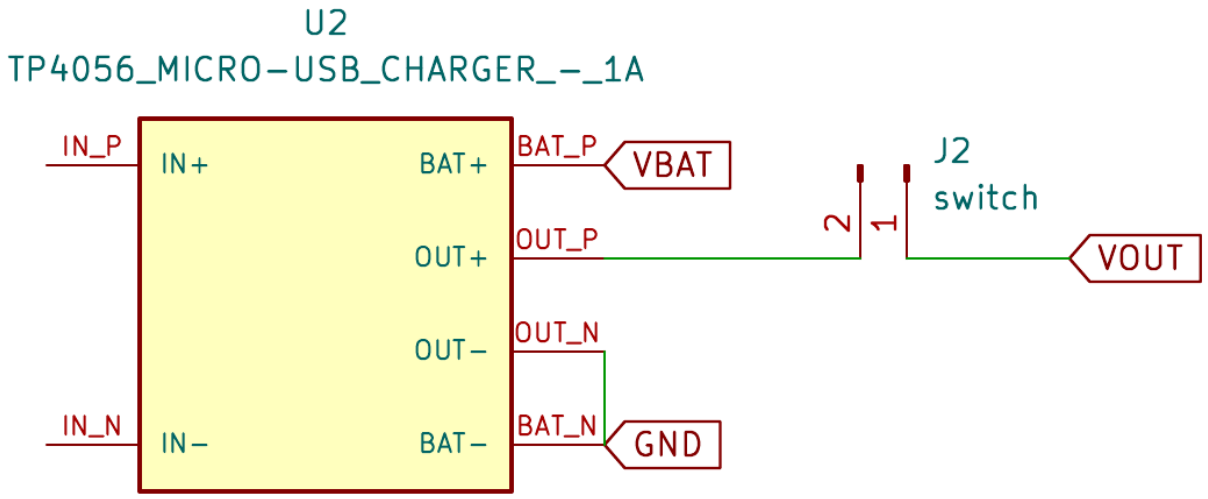


FIG 4.1 TP4056 charger module U2 and power switch J2 (schematic as reviewed).

Charger module · U2

PAD	NET	FUNCTION
B+	VBAT	Cell positive (after reverse protection)
B-	GND	Cell negative
OUT+	OUT_P	Protected output → switch J2 pin 2
OUT-	GND	Protected return — **currently tied to B−**, see finding F2
IN+ / IN-	—	Not used; charging via the module's own USB-C connector

FINDING F2 · CELL PROTECTION BYPASSED

On this module the protection IC switches the **negative** path between `B-` and `OUT-`. With `OUT-`, `B-` and the cell negative all on `GND`, over-discharge, over-current and short-circuit protection are disabled. Give the cell negative its own net (e.g. `BAT_N`) connected only to BT1 N and module `B-`; the board ground becomes module `OUT-`.

Switch and bulk decoupling

J2 is a two-pin header for an external on/off switch in series between module `OUT+` and `VOUT`. When open, only the charger and cell remain energised, so the board can charge while switched off. `VOUT` carries a 100 µF / 16 V bulk capacitor (C1); the motor supply `VDRIV` has its own 100 µF + 100 nF pair (C6, C4) close to the driver.

IDEAL-DIODE STAGE

05 Reverse polarity protection

Two AO3401A P-channel MOSFETs in parallel sit in the positive path between the cell holder and `VBAT`. Compared with a Schottky diode (≈0.35 V drop, 10 % of the cell voltage) the stage drops only tens of millivolts.

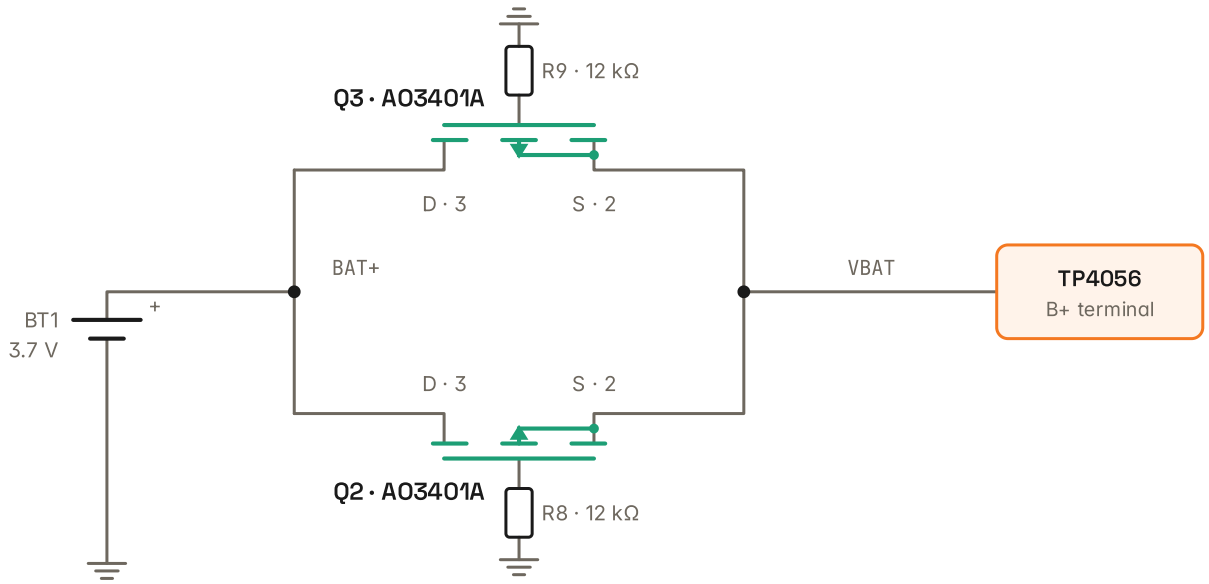


FIG 5.1 Correct topology: drains (pin 3) to the cell, sources (pin 2) to VBAT, each gate (pin 1) pulled to GND by 12 kΩ.

Operation

Cell inserted correctly — the body diode conducts first and lifts the source to ≈3.7 V while the gate sits at 0 V. V_{GS} ≈ −3.7 V fully enhances both FETs, which then bypass the diode.

Cell reversed — the drain is driven to −3.7 V, the body diode is reverse-biased and V_{GS} = 0 V. Both FETs stay off; no current flows.

No DC current flows in R8/R9: the gate is insulated and sits at ground potential. V_{GS} never exceeds 4.2 V, well inside the ±12 V rating, so no Zener clamp is needed.

Losses at 1S

LOAD	DROP	HOTTEST FET
1 A	≈ 40 mV	≈ 25 mW
2 A	≈ 80 mV	≈ 0.10 W
3 A	≈ 120 mV	≈ 0.20 W
4 A	≈ 155 mV	≈ 0.37 W

Worst case: $R_{DS(on)}$ ≈ 65 mΩ per FET at low cell voltage, 60/40 current share.

FINDING F1 · STAGE IS WIRED IN REVERSE

In the current schematic both FETs have **pin 2 (source) on the cell** and **pin 3 (drain) on VBAT**. With a reversed cell the body diodes become forward-biased and current flows backwards through the board — the stage offers no protection. Swap the two connections on Q2 and Q3 as in FIG 5.1.

MX1508 H-BRIDGE

06 Motor driver

Channel 1 of the MX1508 dual H-bridge (U4) drives the motor on J3. Its power stage runs from `VDRIV` — the battery rail after the current shunt — while its logic supply is +3.3 V.

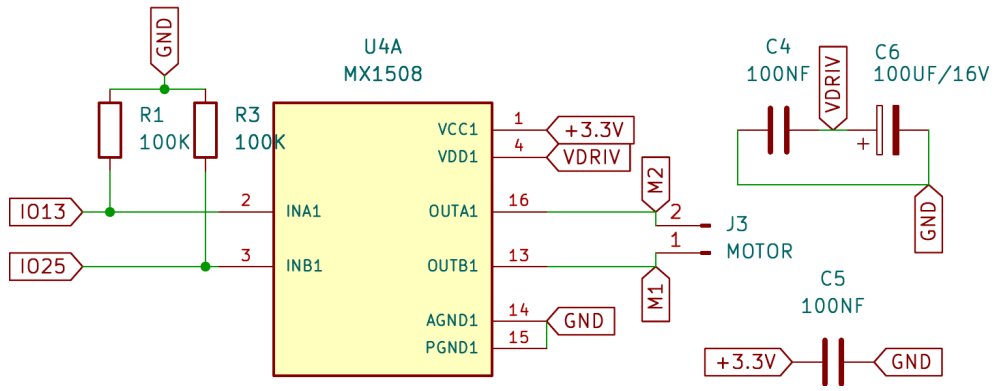


FIG 6.1 MX1508 channel 1 with input pull-downs R1/R3, motor connector J3 and supply decoupling.

Truth table

INA1	INB1	J3-2 / J3-1	MODE
0	0	Z / Z	Coast (standby)
1	0	+ / −	Direction A
0	1	− / +	Direction B
1	1	L / L	Brake

R1 and R3 (100 kΩ) hold both inputs low while the controller boots, so the motor always starts in coast. C7 (100 nF) across the motor terminals suppresses brush noise.

Connections

PIN	NET	ROLE
INA1 · 2	I013	PWM / direction
INB1 · 3	I025	PWM / direction
OUTA1 · 16	M2	J3 pin 2
OUTB1 · 13	M1	J3 pin 1
VDD1/2 · 4, 8	VDRIV	Motor supply
VCC1/2 · 1, 5	+3.3V	Logic supply

Firmware — speed and direction

Arduino-ESP32 3.x

```
constexpr uint32_t PWM_FREQ = 20000;   // 20 kHz: silent; lower it if low-duty torque is weak
constexpr uint8_t  PWM_BITS = 10;      // duty 0…1023

void motorInit() {
  ledcAttach(PIN_MOT_IN1, PWM_FREQ, PWM_BITS);
  ledcAttach(PIN_MOT_IN2, PWM_FREQ, PWM_BITS);
  motorCoast();
}
// speed: -1023 … +1023, sign selects direction
void motorSet(int speed) {
  speed = constrain(speed, -1023, 1023);
  if      (speed > 0) { ledcWrite(PIN_MOT_IN2, 0); ledcWrite(PIN_MOT_IN1,  speed); }
  else if (speed < 0) { ledcWrite(PIN_MOT_IN1, 0); ledcWrite(PIN_MOT_IN2, -speed); }
  else motorCoast();
}
void motorCoast() { ledcWrite(PIN_MOT_IN1, 0);    ledcWrite(PIN_MOT_IN2, 0);    }
void motorBrake() { ledcWrite(PIN_MOT_IN1, 1023); ledcWrite(PIN_MOT_IN2, 1023); }
```

Pin names resolve per module — see the pin map in section 09. Channel 2 of the MX1508 is unused; tie its inputs low (finding F5).

INA219 POWER MONITOR

07 Voltage & current sensing

U5 (INA219) measures the drop across a 10 mΩ shunt R2 placed high-side between `VOUT` and `VDRIV`, and reads the bus voltage on `VDRIV`. The result is the motor's own voltage, current and power — independent of the controller and LED loads.

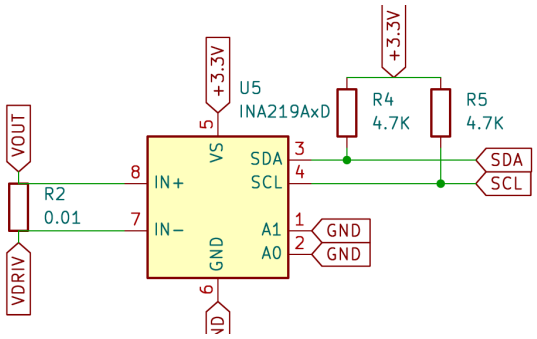


FIG 7.1 INA219 with shunt R2 and I²C pull-ups.

SETTING	VALUE
I²C address	`0x40` (A0 = A1 = GND)
Supply V_S	+3.3 V, C2 100 nF
Shunt R2	10 mΩ, high side
Bus range	16 V (BRNG = 0)
Shunt range	±320 mV (PGA ÷8)
ADC	12-bit, continuous
Pull-ups	R4, R5 4.7 kΩ to 3.3 V

Calibration

Current LSB

100 µA / bit
→ 3.2767 A full scale

CAL register

0.04096 ÷ (100 µA × 0.01 Ω)
= 40960 (0xA000)

Power LSB

20 × Current LSB
= 2 mW / bit

LIBRARY DEFAULT CALIBRATION WILL BE WRONG

Common INA219 libraries assume a 0.1 Ω shunt. With the 10 mΩ shunt on this board their current readings are off by a factor of ten — write the configuration and calibration registers explicitly as below.

```
#include <Wire.h>
constexpr uint8_t INA = 0x40;

void inaWrite(uint8_t reg, uint16_t v) {
  Wire.beginTransmission(INA); Wire.write(reg);
  Wire.write(v >> 8); Wire.write(v & 0xFF); Wire.endTransmission();
}
int16_t inaRead(uint8_t reg) {
  Wire.beginTransmission(INA); Wire.write(reg); Wire.endTransmission(false);
  Wire.requestFrom(INA, (uint8_t)2);
  return (int16_t)((Wire.read() << 8) | Wire.read());
}
void inaInit() {
  Wire.begin(PIN_SDA, PIN_SCL);
  inaWrite(0x00, 0x199F);    // 16 V, PGA ÷8, 12-bit bus + shunt, continuous
  inaWrite(0x05, 40960);     // calibration for 10 mΩ, 100 µA/bit
}
float motorCurrentA() { return inaRead(0x04) * 0.0001f; }                // A
float motorVoltageV() { return ((uint16_t)inaRead(0x02) >> 3) * 0.004f; } // V
float motorPowerW()   { return (uint16_t)inaRead(0x03) * 0.002f; }        // W
```

Shunt dissipation stays small: 22.5 mW at 1.5 A and 90 mW at 3 A. Resolution is 10 µV on the shunt, i.e. 1 mA.

AUXILIARY I/O

08 LED output & encoder input

Switched LED channel · J5

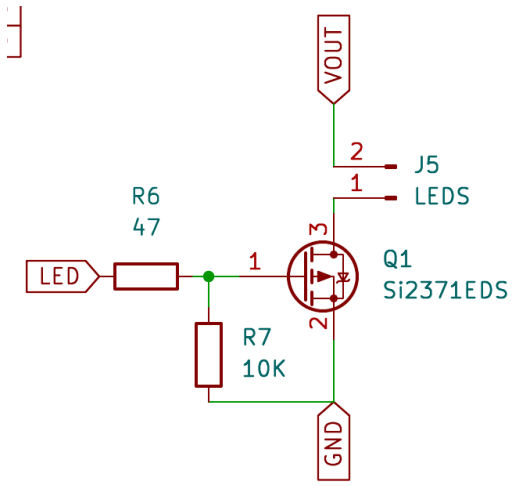


FIG 8.1 LED driver Q1 with gate resistor R6 and pull-down R7.

J5 supplies an external LED load from `VOUT` (pin 2) and switches its return (pin 1) to ground through Q1. R6 (47 Ω) limits gate-charge current from the GPIO; R7 (10 kΩ) keeps the LEDs off while the controller boots.

J5 PIN	NET	ROLE
1	LEDS	LED return (switched)
2	VOUT	LED supply, 3.0–4.2 V

The LED load must include its own current limiting; the channel does not regulate current.

FINDING F3 · Q1 MUST BE N-CHANNEL

Q1 is specified as Si2371EDS, which Vishay lists as a **P-channel** MOSFET. A low-side switch driven from a 3.3 V GPIO needs a logic-level **N-channel** part; with a P-channel device here the LEDs are fed through its body diode and cannot be switched off. Replace with **AO3400A** (same SOT-23 G-S-D pinout, $R_{DS(on)}$ < 48 mΩ at V_{GS} = 2.5 V). Verify against the datasheet before ordering.

```
constexpr uint8_t LED_BITS = 8;
void ledInit()             { ledcAttach(PIN_LED, 5000, LED_BITS); ledcWrite(PIN_LED, 0); }
void ledSet(uint8_t level) { ledcWrite(PIN_LED, level); }    // 0 = off … 255 = full
```

Encoder input · J4

J4 PIN	NET	ROLE
1	GND	Ground
2	IO33	Channel A
3	IO32	Channel B
4	+3.3V	Encoder supply

The encoder is powered from +3.3 V, so its outputs are already at controller logic levels. There are no pull-ups on the board: enable the internal pull-ups for open-collector encoders.

For quadrature decoding, use the ESP32 / ESP32-S3 **pulse counter (PCNT)** peripheral rather than interrupts — it counts in hardware and does not load the CPU at high motor speeds.

ONE FOOTPRINT, TWO CHIPS

09 Controller module compatibility

The TTGO T-Display and T-Display-S3 share the same 2 × 12 header positions. Power and ground land on identical pins; only the GPIO behind each signal pin changes. Each row below is one physical pad on the carrier.

Header 1 · power & motor side

TTGO PIN	TTGO GPIO	S3 PIN	S3 GPIO	NET	FUNCTION
12	3.3V	1	3V3	+3.3V	Logic supply
11	IO36	2	GPIO1	–	Unused
10	IO37	3	GPIO2	–	Unused
9	IO38	4	GPIO3	–	Unused
8	IO39	5	GPIO10	–	Unused
7	**IO32**	6	**GPIO11**	IO32	Encoder B · J4-3
6	**IO33**	7	**GPIO12**	IO33	Encoder A · J4-2
5	**IO25**	8	**GPIO13**	IO25	MX1508 INB1
4	IO26	9	NC	–	Unused
3	IO27	10	NC	–	Unused
2	GND	11	GND	GND	Ground
1	5V	12	5V	VOUT	Module power input

Header 2 · I²C & LED side

TTGO PIN	TTGO GPIO	S3 PIN	S3 GPIO	NET	FUNCTION
13	GND	13	GND	GND	Ground
14	GND	14	GND	GND	Ground
15	**IO21 · SDA**	15	**GPIO43**	SDA	INA219 data
16	**IO22 · SCL**	16	**GPIO44**	SCL	INA219 clock
17	**IO17**	17	**GPIO18**	LED	LED driver Q1 gate
18	IO2	18	GPIO17	–	Unused
19	IO15	19	GPIO21	–	Unused
20	**IO13**	20	**GPIO16**	IO13	MX1508 INA1
21	IO12	21	NC	–	Unused
22	GND	22	GND	GND	Ground
23	GND	23	GND	GND	Ground
24	3.3V	24	3V3	+3.3V	Logic supply

Firmware pin map

Select the module at build time; the rest of the firmware uses functional names only.

```
#if CONFIG_IDF_TARGET_ESP32S3        // LilyGO T-Display-S3
  #define PIN_MOT_IN1  16    // MX1508 INA1
  #define PIN_MOT_IN2  13    // MX1508 INB1
  #define PIN_ENC_A    12    // J4-2
  #define PIN_ENC_B    11    // J4-3
  #define PIN_LED      18
  #define PIN_SDA      43
  #define PIN_SCL      44
#else                                // LilyGO TTGO T-Display (ESP32)
  #define PIN_MOT_IN1  13
  #define PIN_MOT_IN2  25
  #define PIN_ENC_A    33
  #define PIN_ENC_B    32
  #define PIN_LED      17
  #define PIN_SDA      21
  #define PIN_SCL      22
#endif
```

Quick reference

FUNCTION	NET	TTGO T-DISPLAY	T-DISPLAY-S3
Motor IN1	IO13	GPIO13	GPIO16
Motor IN2	IO25	GPIO25	GPIO13
Encoder A	IO33	GPIO33	GPIO12
Encoder B	IO32	GPIO32	GPIO11
LED	LED	GPIO17	GPIO18
I²C SDA	SDA	GPIO21	GPIO43
I²C SCL	SCL	GPIO22	GPIO44

T-DISPLAY-S3 · UART0 ON THE I²C PINS

GPIO43 and GPIO44 are the ESP32-S3's UART0 TX/RX. The boot ROM prints to TX at reset, so the I²C bus sees a short burst of activity before firmware starts — initialise the INA219 after boot. Build with **USB CDC On Boot** enabled so `Serial` uses native USB and never touches these pins.

FINDING F4 · U1 SYMBOL LABELS

In the schematic, U1 (T-Display-S3) pins 6, 7 and 8 carry the labels `IO25`, `IO26`, `IO27`. They must be `IO32`, `IO33`, `IO25` to match the physical pads. As drawn, `IO26`/`IO27` are single-pin nets and `IO25` lands on two different pads.

Resource check

None of the GPIOs used collide with the modules' on-board display, backlight or buttons. On the TTGO T-Display the input-only pins IO36–IO39 are left unused, so every assigned signal supports output and PWM on both modules.

BUILD REFERENCE

10 Connectors & BOM

Connectors

REF	LABEL	PIN 1	PIN 2	PINS 3 / 4
J1	TTGO VIN	GND	VOUT	—
J2	Switch	VOUT	OUT_P (module)	—
J3	Motor	M1 · OUTB1	M2 · OUTA1	—
J4	Encoder	GND	IO33 · A	IO32 · B / +3.3V
J5	LEDs	LEDS (switched)	VOUT	—

Bill of materials

REF	QTY	VALUE / PART	DESCRIPTION
U1	1	TTGO T-Display / T-Display-S3	Controller module, 2 × 12 headers
U2	1	TP4056 module, 1 A	Li-ion charger with protection, USB-C
U4	1	MX1508	Dual H-bridge motor driver (channel 1 used)
U5	1	INA219AxD	I²C current / power monitor
Q2, Q3	2	AO3401A	P-MOSFET, −30 V, SOT-23 — reverse protection
Q1	1	Si2371EDS → **AO3400A**	LED low-side switch, SOT-23 (see F3)
R2	1	10 mΩ	Current-sense shunt
R1, R3	2	100 kΩ	H-bridge input pull-downs
R4, R5	2	4.7 kΩ	I²C pull-ups
R6	1	47 Ω	Q1 gate resistor
R7	1	10 kΩ	Q1 gate pull-down
R8, R9	2	12 kΩ	Reverse-protection gate pull-downs
C1, C6	2	100 µF / 16 V	Bulk: VOUT and VDRIV
C3	1	10 µF	+3.3 V bulk
C2, C4, C5, C7	4	100 nF	Decoupling; C7 across motor
BT1	1	Keystone 1042P	18650 cell holder
J1–J3, J5	4	2-pin header	VIN, switch, motor, LEDs
J4	1	4-pin header	Encoder

BEFORE FABRICATION

11 Design review

Findings from the Rev A schematic review. F1–F3 are blocking: each one either removes a protection function or prevents a feature from working. Resolve them before ordering boards.

ID	SEVERITY	FINDING	REQUIRED CHANGE
F1	Blocking	Reverse-protection FETs Q2/Q3 wired source-to-cell, drain-to-VBAT; body diodes conduct with a reversed cell.	Pin 3 (drain) → BT1 P, pin 2 (source) → VBAT on both FETs.
F2	Blocking	TP4056 module `OUT-` tied to `B-`; cell protection bypassed.	New net `BAT_N` for BT1 N and module `B-` only; board GND = `OUT-`.
F3	Blocking	Q1 Si2371EDS is P-channel in a low-side position; LEDs cannot be turned off.	Use a logic-level N-channel FET, e.g. AO3400A (same pinout).
F4	Major	U1 (T-Display-S3) pins 6–8 labelled IO25/IO26/IO27.	Relabel to IO32 / IO33 / IO25 (section 09).
F5	Minor	MX1508 channel 2 inputs INA2/INB2 (pins 6, 7) left floating.	Tie both to GND.
F6	Advisory	Module fed with VOUT (3.0–4.2 V) on its 5 V pin and via J1.	Measure the module's 3.3 V rail at a 3.3 V cell; use a single power entry path.
F7	Advisory	Net names tied to TTGO GPIO numbers (IO13, IO25…).	Rename by function: MOT_IN1, MOT_IN2, ENC_A, ENC_B, LED_CTRL.

Bring-up checklist

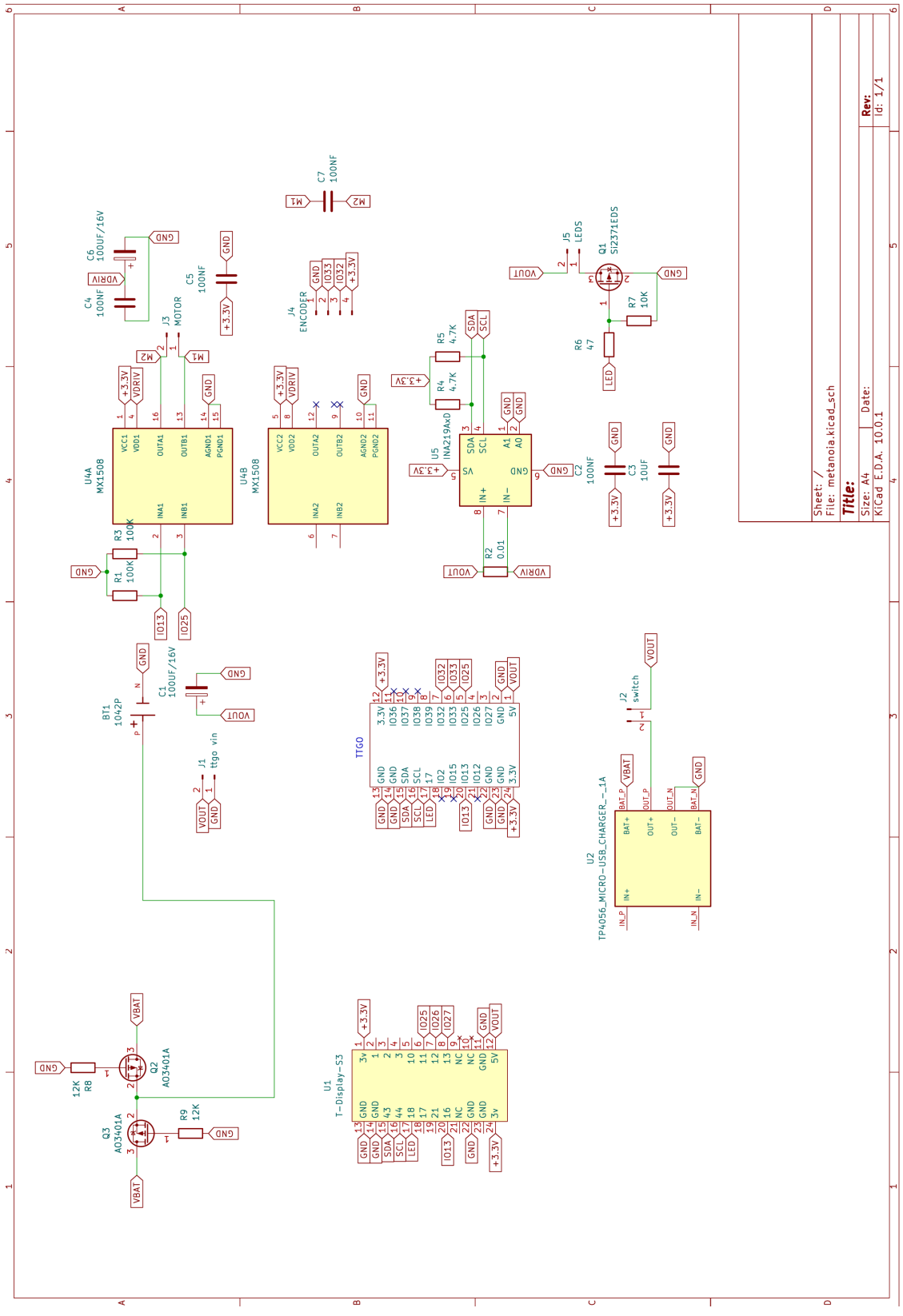
- Power from a bench supply at 3.7 V, 100 mA limit, cell holder empty.
- Apply −3.7 V to the holder terminals: supply current must read ≈ 0 mA (F1).
- Apply +3.7 V: measure BAT+ → VBAT drop, < 0.1 V at 2 A.
- Short OUT+ to OUT− briefly: the module must cut the output (F2).
- Confirm +3.3 V on the module 3V pins across 3.3–4.2 V input.
- I²C scan must return `0x40`.
- Motor coasts at boot; test both directions and brake at 25 % duty.
- Compare INA219 current with a DMM in series with the motor.

VERIFIED IN THIS REVIEW

Controller header pin compatibility (both modules), power-pin alignment, I²C pull-ups, INA219 addressing and shunt placement, H-bridge boot-state pull-downs, and reverse-protection V_{GS} margin at 1S.

AS REVIEWED · REV A

A Schematic



Sheet rotated 90°. Shown before the changes listed in section 11.

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